

PR_01.3

Nota: Las capturas pueden variar en color, resolución y aspecto debido al uso de dos equipos diferentes durante la práctica.

Bloque 1: Comandos de Información y Preparación

Bloque 3: Permisos y Propiedad de Archivos

Bloque 4: Gestión de Servicios con `systemctl`

Bloque 5: Gestión de `ufw`

Bloque 1: Comandos de Información y Preparación

1. Identidad del Usuario:

```
21 servidor in ~  
◆ → id  
uid=1000(akoarguel) gid=1000(akoarguel) groups=1000(akoarguel),  
)  
21 servidor in ~  
◆ →
```

2. Usuarios conectados:

```
21 servidor in ~  
◆ → who  
akoarguel tty1          2025-10-23 21:09  
  
21 servidor in ~  
◆ → _
```

3. Historial de Conexiones

```
21 servidor in ~  
◆ → last  
reboot    system boot    6.8.0-86-generic Thu Oct 23 21:08    still running  
reboot    system boot    6.8.0-86-generic Thu Oct 23 20:33    still running  
reboot    system boot    6.8.0-85-generic Thu Oct 23 20:19 - 20:33    (00:13)  
reboot    system boot    6.8.0-85-generic Wed Oct 22 19:58 - 20:33    (1+00:34)  
reboot    system boot    6.8.0-85-generic Wed Oct 22 17:00 - 20:33    (1+03:32)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 19:07 - 20:33    (3+01:26)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 18:41 - 19:07    (00:25)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 18:16 - 19:07    (00:50)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 18:14 - 19:07    (00:52)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 17:24 - 19:07    (01:42)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 17:07 - 19:07    (01:59)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 17:02 - 19:07    (02:04)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 16:23 - 19:07    (02:44)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 16:03 - 19:07    (03:03)  
reboot    system boot    6.8.0-85-generic Mon Oct 20 15:57 - 16:03    (00:05)  
reboot    system boot    6.8.0-85-generic Thu Oct 16 15:17 - 16:03    (4+00:46)  
reboot    system boot    6.8.0-85-generic Thu Oct 16 15:10 - 16:03    (4+00:53)  
  
wtmp begins Thu Oct 16 15:10:30 2025  
  
21 servidor in ~  
◆ → _
```

4. Crear Entorno de Trabajo:

```
21 servidor in ~  
◆ → mkdir ~/practicas_linux  
mkdir: created directory '/home/akoarguel/practicas_linux'  
  
21 servidor in ~  
◆ →
```

5. Estructura de Directorios:

```
21 servidor in ~/practicas_linux  
◆ → pwd  
/home/akoarguel/practicas_linux  
  
21 servidor in ~/practicas_linux  
◆ → mkdir proyectos documentos scripts  
mkdir: created directory 'proyectos'  
mkdir: created directory 'documentos'  
mkdir: created directory 'scripts'  
  
21 servidor in ~/practicas_linux  
◆ →
```

Bloque 2: Gestión de Usuarios y Grupos

1. Crear Grupos

```
21 servidor in ~/practicas_linux  
◆ → sudo groupadd analistas  
  
21 servidor in ~/practicas_linux  
◆ → sudo groupadd becarios  
  
21 servidor in ~/practicas_linux  
◆ → sudo groupadd desarrolladores_
```

2. Verificar grupos

```
21 servidor in ~/practicas_linux  
◆ → grep -E 'desarrolladores|analistas|becarios' /etc/group  
desarrolladores:x:1001:  
analistas:x:1002:  
becarios:x:1003:  
  
21 servidor in ~/practicas_linux  
◆ →
```

3. Crear un Usuario Básico

```

21 servidor in ~
◆ → sudo adduser juan
[sudo] password for akoarguel:
info: Adding user `juan' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `juan' (1004) ...
info: Adding new user `juan' (1004) with group `juan (1004)' ...
info: Creating home directory `/home/juan' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for juan
Enter the new value, or press ENTER for the default
    Full Name []: juan
    Room Number []: 1004
    Work Phone []: 619424033
    Home Phone []: Burgos
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user `juan' to supplemental / extra groups `users' ...
info: Adding user `juan' to group `users' ...

21 servidor in ~
◆ →

```

4. Crear Usuario con Grupo Primario

```

21 servidor in ~
◆ → sudo adduser --ingroup desarrolladores ana
info: Adding user `ana' ...
info: Selecting UID from range 1000 to 59999 ...

info: Adding new user `ana' (1001) with group `desarrolladores (1001)' ...
info: Creating home directory `/home/ana' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for ana
Enter the new value, or press ENTER for the default
    Full Name []: Ana
    Room Number []: 1005
    Work Phone []: 60607745
    Home Phone []:
    Other []:
Is the information correct? [Y/n] Y
info: Adding new user `ana' to supplemental / extra groups `users' ...
info: Adding user `ana' to group `users' ...

21 servidor in ~
◆ → _

```

5. Crear Usuario Completo:

(Otra forma de crearlo)

```

21 servidor in ~
◆ → sudo useradd -m -g analistas -G desarrolladores,becarios david

21 servidor in ~
◆ → sudo passwd david
New password:
Retype new password:
passwd: password updated successfully

21 servidor in ~
◆ →

```

6. Establecer contraseñas

Ya se ha realizado en los ejercicios anteriores.

7. Verificar Usuarios:

```
21 servidor in ~  
♦ → tail -n 3 /etc/passwd  
juan:x:1004:1004:juan,1004,619424033,Burgos:/home/juan:/bin/bash  
ana:x:1001:1001:Ana,1005,60607745,:/home/ana:/bin/bash  
david:x:1005:1002::/home/david:/bin/sh  
  
21 servidor in ~  
♦ →
```

8. Cambiar de Usuario:

```
21 servidor in ~  
♦ → su - juan  
Password:  
juan@servidor:~$ whoami  
juan  
juan@servidor:~$ pwd  
/home/juan  
juan@servidor:~$ _
```

9. Modificar Grupos de un Usuario:

```
21 servidor in ~  
♦ → sudo usermod -g becarios -a -G analistas juan  
  
21 servidor in ~  
♦ →
```

10. Verificar modificación

```
21 servidor in ~  
♦ → id juan  
uid=1004(juan) gid=1003(becarios) groups=1003(becarios),100(users),1002(analistas)  
  
21 servidor in ~  
♦ →
```

11. Bloquear una Cuenta

```
21 servidor in ~  
♦ → sudo usermod -L juan  
  
21 servidor in ~  
♦ →
```

12. Intentar cambiar a usuario bloqueado:

```
21 servidor in ~  
♦ → su - juan  
Password:  
su: Authentication failure  
  
21 servidor in ~  
♦ →
```

13. Desbloquear una Cuenta:

```
21 servidor in ~  
◆ → sudo usermod -U juan  
  
21 servidor in ~  
◆ → su - juan  
Password:  
juan@servidor:~$ _
```

14. Eliminar un Grupo:

El comando fallará. El SO no te permite eliminar un grupo si un usuario lo tiene configurado como su grupo primario. Para poder borrarlo, primero tendrías que cambiar el grupo de juan.

```
21 servidor in ~  
◆ → sudo groupdel becarios  
groupdel: cannot remove the primary group of user 'juan'  
  
21 servidor in ~  
◆ → _
```

15. Eliminar Usuario y su Directorio:

```
21 servidor in ~  
◆ → sudo deluser --remove-home juan  
info: Looking for files to backup/remove ...  
info: Removing files ...  
info: Removing crontab ...  
info: Removing user 'juan' ...  
userdel: group juan not removed because it is not the primary group of user juan.  
  
21 servidor in ~  
◆ →
```

Bloque 3: Permisos y Propiedad de Archivos

1. Crear Archivos de Prueba:

```
21 servidor in ~/practicas_linux  
◆ → touch proyectos/informe.txt  
  
21 servidor in ~/practicas_linux  
◆ → touch scripts/lanzar_app.sh
```

```
.  
├── documentos  
├── proyectos  
│   └── informe.txt  
└── scripts  
    └── lanzar_app.sh
```

2. Ver permisos:

```

21 servidor in ~/practicas_linux
♦ → ls -lR
.:
total 12
drwxrwxr-x 2 akoarguel akoarguel 4096 oct 23 21:12 documentos
drwxrwxr-x 2 akoarguel akoarguel 4096 oct 23 21:37 proyectos
drwxrwxr-x 2 akoarguel akoarguel 4096 oct 23 21:38 scripts

./documentos:
total 0

./proyectos:
total 0
-rw-rw-r-- 1 akoarguel akoarguel 0 oct 23 21:37 informe.txt

./scripts:
total 0
-rw-rw-r-- 1 akoarguel akoarguel 0 oct 23 21:38 lanzar_app.sh

```

3. Cambiar propietario:

```

21 servidor in ~/practicas_linux
♦ → sudo chown ana proyectos/informe.txt

21 servidor in ~/practicas_linux
♦ → _

```

4. Cambiar Grupo

```

21 servidor in ~/practicas_linux
♦ → sudo chgrp desarrolladores proyectos

21 servidor in ~/practicas_linux
♦ → _

```

5. Cambiar Propietario y Grupo

```

21 servidor in ~/practicas_linux
♦ → sudo chown david:analistas scripts/lanzar_app.sh

21 servidor in ~/practicas_linux
♦ → _

```

6. Permisos con Notación Octal (Archivo):

Significado:

- `u=rw` → el propietario puede leer y escribir
- `g=r` → el grupo solo puede leer
- `o=` → otros sin permisos

```

21 servidor in ~/practicas_linux
♦ → sudo chown ana:desarrolladores proyectos/informe.txt

21 servidor in ~/practicas_linux
♦ → sudo chmod 640 proyectos/informe.txt

```

7. Permisos con Notación Octal (Directorio):

Significado:

- **Propietario:** Leer (4) + Escribir (2) + Ejecutar (1) = 7
- **Grupo:** Leer (4) + Ejecutar (1) = 5
- **Otros:** (No se especifica, asumimos 0) = 0

```
21 servidor in ~/practicas_linux
◆ → chmod 750 documentos

21 servidor in ~/practicas_linux
◆ →
```

8. Verificar Permisos:

```
21 servidor in ~/practicas_linux
◆ → ls -lR
.:
total 12
drwxr-x--- 2 akoarguel akoarguel      4096 oct 23 21:12 documentos
drwxrwxr-x 2 akoarguel desarrolladores 4096 oct 23 21:37 proyectos
drwxrwxr-x 2 akoarguel akoarguel      4096 oct 23 21:38 scripts

./documentos:
total 0

./proyectos:
total 0
-rw-r----- 1 ana desarrolladores 0 oct 23 21:37 informe.txt

./scripts:
total 0
-rw-rw-r-- 1 david analistas 0 oct 23 21:38 lanzar_app.sh
```

9. Permisos con Notación Simbólica (Añadir):

```
21 servidor in ~/practicas_linux
◆ → sudo chmod u+x scripts/lanzar_app.sh

21 servidor in ~/practicas_linux
◆ →
```

10. Permisos con Notación Simbólica (Quitar):

```
21 servidor in ~/practicas_linux
◆ → chmod o-r proyectos

21 servidor in ~/practicas_linux
◆ →
```

11. Permisos Recursivos:

```
21 servidor in ~/practicas_linux
◆ → mkdir proyectos/version2
mkdir: created directory 'proyectos/version2'

21 servidor in ~/practicas_linux
◆ → touch proyectos/version2/notas.txt

21 servidor in ~/practicas_linux
◆ → sudo chown -R david proyectos
```

12. Permiso Especial SGID de Directorio:

```
21 servidor in ~/practicas_linux  
♦ → chmod 2750 documentos_
```

```
21 servidor in ~/practicas_linux  
♦ → su - david  
Password:  
$ _
```

13. Comprobar `umask`:

```
♦ akoarguel at servidor in ~ in  
└─λ umask  
0002  
♦ akoarguel at servidor in ~ in  
└─λ
```

14. Efecto de `umask`:

```
♦ akoarguel at servidor in ~ in  
└─λ umask 077  
♦ akoarguel at servidor in ~ in  
└─λ umask  
0077  
♦ akoarguel at servidor in ~ in  
└─λ
```

```
♦ akoarguel at servidor in ~ in  
└─λ pwd  
/home/akoarguel  
♦ akoarguel at servidor in ~ in  
└─λ touch privado.txt  
♦ akoarguel at servidor in ~ in  
└─λ _
```

```
♦ akoarguel at servidor in ~ in  
└─λ ls -l  
total 8  
drwxrwxr-x 5 akoarguel akoarguel 4096 oct 21 07:44 PR_01.1  
drwxrwxr-x 3 akoarguel akoarguel 4096 oct 27 07:50 PR_01.2  
-rw----- 1 akoarguel akoarguel    0 oct 30 08:44 privado.txt  
♦ akoarguel at servidor in ~ in  
└─λ _
```

```
♦ akoarguel at servidor in ~ in  
└─λ umask 02  
♦ akoarguel at servidor in ~ in  
└─λ umask  
0002  
♦ akoarguel at servidor in ~ in  
└─λ
```

Bloque 4: Gestión de Servicios con `systemctl`

1. Estado Detallado de un Servicio:

Usamos `sudo apt update` y luego instalamos cups con `sudo apt install cups`.

Ahora comprobamos el estado del servicio con `systemctl status cups`.

```
akoarguel at servidor in ~ in
λ sudo systemctl status cups
• cups.service - CUPS Scheduler
   Loaded: loaded (/usr/lib/systemd/system/cups.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-10-30 08:12:25 UTC; 1min 12s ago
   TriggeredBy: • cups.socket
                • cups.path
   Docs: man:cupsd(8)
   Main PID: 3520 (cupsd)
   Status: "Scheduler is running..."
   Tasks: 1 (limit: 2264)
   Memory: 2.8M (peak: 16.3M)
   CPU: 817ms
   CGroup: /system.slice/cups.service
           └─3520 /usr/sbin/cupsd -l

oct 30 08:12:25 servidor systemd[1]: Starting cups.service - CUPS Scheduler...
oct 30 08:12:25 servidor systemd[1]: Started cups.service - CUPS Scheduler.
```

El servicio está activo.

2. Comprobación Rápida:

Utilizamos el comando `systemctl is-active cups`.

```
akoarguel at servidor in ~ in
λ systemctl is-active cups
active
akoarguel at servidor in ~ in
λ
```

3. Ver Archivo de Unidad:

```
akoarguel at servidor in ~ in
λ cat cups.service
cat: cups.service: No such file or directory
akoarguel at servidor in ~ in
λ systemctl cat cups.service
# /usr/lib/systemd/system/cups.service
[Unit]
Description=CUPS Scheduler
Documentation=man:cupsd(8)
After=network.target nss-user-lookup.target nslcd.service
Requires=cups.socket

[Service]
ExecStart=/usr/sbin/cupsd -l
Type=notify
Restart=on-failure

[Install]
Also=cups.socket cups.path
WantedBy=printer.target multi-user.target
akoarguel at servidor in ~ in
λ _
```

4. Detener un Servicio:

```

└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ systemctl stop cups
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to stop 'cups.service'.
Authenticating as: akoarguel
Password:
==== AUTHENTICATION COMPLETE ====
└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ

```

Nos solicitará la contraseña de administrador para parar `cups.service`.

5. Iniciar un Servicio:

```

└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ systemctl start cups
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to start 'cups.service'.
Authenticating as: akoarguel
Password:
==== AUTHENTICATION COMPLETE ====
└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ _

```

6. Reiniciar un Servicio:

```

└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ systemctl restart cups
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to restart 'cups.service'.
Authenticating as: akoarguel
Password:
==== AUTHENTICATION COMPLETE ====
└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ _

```

7. Habilitar para el Arranque:

```

└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ sudo systemctl enable cups
Synchronizing state of cups.service with SysV service script with
Executing: /usr/lib/systemd/systemd-sysv-install enable cups
└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ _

```

8. Verificar si está Habilitado:

```

└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ systemctl is-enabled cups
enabled
└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ

```

9. Deshabilitar para el Arranque:

```

└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ sudo systemctl disable cups
Synchronizing state of cups.service with SysV service script with /usr/lib/systemd/sys
Executing: /usr/lib/systemd/systemd-sysv-install disable cups
Removed "/etc/systemd/system/sockets.target.wants/cups.socket".
Removed "/etc/systemd/system/printer.target.wants/cups.service".
Removed "/etc/systemd/system/multi-user.target.wants/cups.path".
Removed "/etc/systemd/system/multi-user.target.wants/cups.service".
Disabling 'cups.service', but its triggering units are still active:
cups.socket
└─◆ akoarguel at ◆ servidor in ◆ ~ in
└─λ

```

10. Enmascarar un Servicio:

```
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ systemctl mask cups
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-unit-files ====
Authentication is required to manage system service or unit files.
Authenticating as: akoarguel
Password:
==== AUTHENTICATION COMPLETE ====
Created symlink /etc/systemd/system/cups.service → /dev/null.
Masking 'cups.service', but its triggering units are still active:
cups.socket
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ _
```

Bloque 5: Gestión de **ufw**

1. Comprobar Estado y Activar UFW:

```
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ sudo ufw status
Status: inactive
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ _
```

```
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ sudo ufw enable
Firewall is active and enabled on system startup
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ _
```

2. Permitir un Servicio Web (HTTP):

```
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ sudo ufw allow http
Rule added
Rule added (v6)
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ
```

3. Abrir un Puerto Específico:

```
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ sudo ufw allow 8080/tcp
Rule added
Rule added (v6)
❖ akoarguel at ❖ servidor in ❖ ~ in
❖ λ _
```

```

└─$ sudo ufw status
Status: active

To Action From
--
80/tcp ALLOW Anywhere
8080/tcp ALLOW Anywhere
80/tcp (v6) ALLOW Anywhere (v6)
8080/tcp (v6) ALLOW Anywhere (v6)

```

4. Permitir un Rango de Puertos:

```

└─$ sudo ufw allow 3000:3100/tcp
Rule added
Rule added (v6)
└─$

```

5. Bloquear una Dirección IP:

```

└─$ sudo ufw deny from 192.168.100.50
Rule added
└─$

```

6. Listar Reglas para Borrar:

```

└─$ sudo ufw delete 2
Deleting:
allow 8080/tcp
Proceed with operation (y/n)? y
Rule deleted
└─$

```

7. Eliminar una Regla:

```

Status: active

To Action From
--
[ 1] 80/tcp ALLOW IN Anywhere
[ 2] 3000:3100/tcp ALLOW IN Anywhere
[ 3] Anywhere DENY IN 192.168.100.50
[ 4] 80/tcp (v6) ALLOW IN Anywhere (v6)
[ 5] 8080/tcp (v6) ALLOW IN Anywhere (v6)
[ 6] 3000:3100/tcp (v6) ALLOW IN Anywhere (v6)

```